ML Interview Revision

Decision Trees + Random Forest

Quick revision sheet based on the questions and explanations we covered. Read the shortcut section when you only have a few minutes.

■ Quick Shortcut Revision

Decision Tree: A tree repeatedly splits data into smaller groups so that the groups become more pure.

Gini Impurity: Measures class disorder. **0 = completely pure**. For binary classification, **0.5 = maximum impurity**. Formula: **Gini = 1** – Σp^2.

Best Split: Try different feature + threshold combinations and choose the one with the **lowest weighted child impurity**.

Continuous Features: Sort the feature values and test thresholds between adjacent values, usually using their midpoint.

Overfitting: A very deep tree can memorize individual training samples instead of learning general patterns.

Stopping Criteria: Common controls are **max_depth**, **min_samples_split**, and **min_samples_leaf**.

Regression Tree: Classification uses criteria such as Gini/entropy. Regression commonly uses variance, MSE, or squared error.

Random Forest: Many decision trees trained using **bootstrap samples** and **random feature subsets**, then combined.

Bagging: Bootstrap Aggregating: train multiple models on bootstrap samples and combine their predictions.

OOB Error: Samples not selected for a particular bootstrap dataset are **out-of-bag** and can be used to estimate generalization error.

Why Random Forest Reduces Variance: Trees are diverse and less correlated. Their errors partially cancel when their predictions are averaged or voted.

■ Decision Trees — Questions We Did

1. What is a Decision Tree?

A tree-based model that makes predictions through a sequence of feature-based splits. Internal nodes perform decisions/splits and leaf nodes make the final prediction.

2. What is Gini impurity?

Gini impurity measures how mixed or disordered the classes are in a node. A completely pure node has Gini = 0.

3. What is the Gini formula?

Gini = 1 – Σ $\mathbf{p_k}^{\mathbf{2}}$, where p_k is the proportion of samples belonging to class k.

4. What does p_k mean?

It is the proportion/probability of class k in the current node.

5. What is Gini for a 20% / 80% split?

Gini = $1 - 0.2^2 - 0.8^2$ = **0.32**.

6. What is Gini for a 50% / 50% split?

Gini = $1 - 0.5^2 - 0.5^2$ = **0.5**. This is the maximum impurity for binary classification.

7. How does a Decision Tree choose the best split?

It tests candidate feature + threshold combinations, calculates the weighted impurity after each split, and chooses the split with the lowest weighted impurity.

8. Why do we weight child impurities?

Because larger groups should have more influence on the final split score. A group containing 95 samples should matter more than a group containing 5.

9. What is the weighted Gini formula?

$Gini_{split} = (N_L/N)Gini_L + (N_R/N)Gini_R$.

10. Can the same feature be used multiple times?

Yes. The same feature can be reused later with a different threshold inside a child node.

11. How are thresholds selected for continuous variables?

Sort the unique values and consider thresholds between adjacent values. A midpoint is commonly used.

12. Why don't we test every possible real number as a threshold?

Many different thresholds produce the exact same partition of the training samples. The partition matters, not the exact numerical value of the threshold.

13. When does a Decision Tree stop splitting?

It may stop when the node is pure, max_depth is reached, minimum-sample constraints prevent a split, or no useful/valid split remains.

14. Why can a deep Decision Tree overfit?

A deep tree can keep splitting until it isolates individual training examples, effectively memorizing the training data.

15. How does max_depth help?

It limits how many levels the tree can grow, preventing the tree from becoming excessively complex.

16. How does min_samples_leaf help?

It requires every leaf to contain at least a certain number of samples, preventing tiny leaves that can memorize individual examples.

17. How does a Regression Tree differ?

Classification trees use class-purity measures such as Gini or entropy. Regression trees use target-value measures such as variance or MSE.

18. What is the weighted variance idea?

The tree chooses the split that minimizes weighted child variance: $\mathbf{(N_L/N)Var(y_L) + (N_R/N)Var(y_R)}$.

19. Advantages of Decision Trees?

Easy to interpret, require little preprocessing, usually do not require feature scaling, and can capture nonlinear relationships.

20. Disadvantages of Decision Trees?

They can overfit easily, have high variance, can be unstable to small changes in training data, and use a greedy strategy that does not guarantee a globally optimal tree.

■ Random Forest — Questions We Did

1. What is Random Forest?

An ensemble of many Decision Trees. Each tree uses a bootstrap sample of the training data and random subsets of features. Their predictions are then combined.

2. What is bootstrap sampling?

Sampling training examples **with replacement**. Therefore, some samples can appear multiple times while some samples may not appear at all in a particular tree's dataset.

3. What is Bagging?

Bagging stands for **Bootstrap Aggregating**. Multiple models are trained on bootstrap datasets and their predictions are aggregated.

4. How is Random Forest related to Bagging?

Random Forest uses the bagging idea with Decision Trees, while also randomly selecting a subset of features at each split.

5. Why randomly select features?

To create diversity and reduce correlation between trees. If every tree always used the same features, the trees could become too similar.

6. How are Random Forest predictions combined?

For classification, use **majority voting**. For regression, usually use the **average** of the tree predictions.

7. Why does Random Forest usually overfit less than one Decision Tree?

Individual trees can have high variance, but the forest combines many diverse, less-correlated trees. Their errors partially cancel when predictions are averaged or voted.

8. What is variance?

Variance describes how much the learned model or its predictions change when the training data changes.

9. Does Random Forest make every tree low variance?

No. Individual trees can still have high variance. The ensemble reduces the overall variance through aggregation.

10. What is OOB error?

OOB means **Out-of-Bag**. For each tree, the training samples that were not selected in its bootstrap sample are OOB for that tree. Their predictions can be used to estimate generalization error.

11. Does OOB error require a separate validation set?

No. OOB samples naturally provide an internal estimate of generalization performance.

12. What is the key explanation for Random Forest variance reduction?

Random Forest reduces variance by averaging/voting across many diverse, less-correlated Decision Trees. Individual trees can have high variance, but their errors partially cancel when combined.

■ Interview Cheat Sheet

- **Decision Tree:** A model that recursively splits data to create increasingly pure groups.
- **Gini:** Measures class impurity. 0 means completely pure; 0.5 is maximum for binary classification.
- **Best Split:** Choose the feature and threshold that produce the lowest weighted child impurity.
- **Overfitting:** Deep trees can memorize training samples. Control complexity using max_depth and minimum-sample constraints.
- **Regression Tree:** Uses variance/MSE/squared error rather than classification impurity such as Gini.
- **Bagging:** Bootstrap samples + train multiple models + aggregate predictions.
- **Random Forest:** Bagging of Decision Trees + random feature selection.
- **Variance:** How sensitive the learned model is to changes in the training data.
- **Random Forest Advantage:** Many diverse, less-correlated trees reduce variance because their errors partially cancel.
- **OOB:** Training samples left out of a tree's bootstrap sample can estimate generalization error.